BEL Circle surface junction redesign

Revision R1 - feasibility design basis - 22 October 2026

This package develops a dimensioned, editable candidate for the two surface intersections around the Kuvempu Underpass and MS Ramaiah Road / BEL Road corridor in Bengaluru. It preserves the underpass alignment and separates through traffic, turning traffic and pedestrian movement in the surface concept. It is a feasibility study, without a percentage-design designation, capacity prediction or construction approval.

The principal finding is a space trade-off. Two continuous surface through lanes in each direction, dedicated turning pockets and connected footpaths require widening or reallocation beyond parts of the visible paved footprint. Candidate A shows that requirement explicitly. It is not demonstrated to fit the public road reserve. An alternative with fewer surface through lanes or restricted turns must be tested if the footprint, structure or demand evidence rejects this arrangement.

1. Scope and deliverables

The seven drawing sheets cover existing conditions and conflicts, the proposed lane plan, movement and signal stages, walking and bus access, footprint and future structure constraints, lane continuity with dimensions, and an enlarged junction detail. The package includes one layered DXF, one RFC 7946 GeoJSON, individual 6000 x 4200 PNG sheets, editable SVG sheets, a combined drawing PDF, this editable basis and its PDF, movement and crossing registers, a conflict screening matrix, coordinate parameters, source copies and a reproducible generator.

The DXF contains 2D polylines and labels in metres, not a Civil 3D corridor or a finished alignment model. The GeoJSON contains longitude/latitude geometry with persistent feature IDs. Neither file is a calibrated simulation network. No traffic volumes, queue demands, saturation flows, signal splits, costs or benefit percentages have been invented.

The supplied CityWays drawing and earlier notebook sketch are evidence of the author's intent. Arrows, lane counts and annotations inside them are not independent survey facts or instructions that override the user's request. Their useful principles are movement separation, continuity, closure of unused pavement and future transit provision. Their depicted lane quantities and structures are not adopted as established requirements.

2. Source evidence and existing reconstruction

SAT-01 is the supplied screenshot, 2026-10-21-222429_hyprshot.png. It shows the two open underpass approaches, portal edges, the broad surface crossing area, surface slip movements and vehicles. Its screenshot date does not establish the satellite imagery date. Trees, labels, shadows and image resolution obscure some boundaries. A single image cannot establish traffic volumes, queue distributions, effective lane use, signal phases or pedestrian crossing behaviour.

OSM-01 is the previously supplied bel-circle-osm.xml. It maps the upper surface link west-to-east, the lower surface link east-to-west, separate crossroad directions, and the ORR carriageways at layer -1. Several surface links have a two-lane tag. The westbound departure link and crossroad links do not establish complete lane counts. The underpass links carry three-lane tags per direction. Those are mapped attributes, requiring field confirmation; a lane tag is not an effective discharge-lane count.

The four central surface nodes are 11554370602 and 11554370603 at the northern crossing, and 11554370601 and 11554370600 at the southern crossing. Taking the mean of the paired carriageway nodes at each end gives a mapped axis separation of 36.544 m. Reporting this as about 36.5 m is appropriate to the source. It is not surveyed signal-to-signal spacing, usable queue storage or deck length.

OSM-02 is a current API extract requested during this review. The archived comparison records coordinate differences for the eight control nodes used here. This checks whether the chosen mapping controls changed; it

does not establish that the whole OSM network or the on-site layout is current. OSM attribution and the source XML licence information are preserved.

The user's descriptions of widening approaches, narrowing exits, weaving, poor walking continuity and concentrated queues are recorded as reported operating problems. The image supports the existence of substantial open pavement and turning opportunities. The seriousness, frequency and causal contribution of each reported problem still need observation. Sheet 01 therefore labels its red trajectories as conflict hypotheses rather than observed vehicle tracks or crash locations.

3. Coordinate basis and trace accuracy

CAD uses WGS 84 / UTM zone 43N, EPSG:32643, in metres. The local drawing origin is the midpoint of the two OSM junction averages. The local longitudinal axis follows their connecting line. The perpendicular axis completes an orthonormal frame. The exact UTM origin and transformation are in design_parameters.json. Local drawing coordinates are not a separate survey datum and must not be assigned directly to EPSG:32643 without applying that transform.

Four interpreted image points at the carriageway centres near the tunnel mouths are matched to OSM nodes 3830966864, 3830928511, 3830966859 and 3830928535. An affine fit relates image pixels to local metres. The selected pixels and residuals are saved. The very small fit residual describes agreement between those selected points only. The points are clustered and partially dependent; there is no independent accuracy check. It must never be presented as centimetre positioning accuracy.

Visible surface edges and open trenches are manually traced from the screenshot using that transform. They are interpretation layers. They are not property lines, road-reserve boundaries, engineering kerbs, cadastral features or structural as-builts. A discrepancy between OSM centrelines and the interpreted image is retained as uncertainty instead of being concealed by claiming one is exact. No satellite-derived width is treated as a certified existing dimension.

The source register includes file hashes and archived originals. The editable geometry carries MAPPED, INTERPRETED or ASSUMED evidence labels. The original generator and original concept files remain intact in the parent directory; they are superseded for this review by R1, not silently overwritten.

4. Critical review of the earlier concept

The old plan labelled the arrangement a 30% tight urban diamond concept and assumed 45 m between terminals. Neither the maturity designation nor a final interchange type was established. R1 uses a coordinated compound surface junction description, retains the measured OSM reference spacing and separately identifies proposed axis shifts.

The previous north-south corridor was straightened through a site with curved approaches. Its solid median and kerb lines extended through turning spaces. Stop lines spanned both traffic directions, several crossroad turning movements were absent, and some drawing lines represented lane centres while appearing to be lane divisions. R1 gives every proposed movement an ID, uses direction-specific stop bars and excludes the lane envelopes from raised islands.

The old plan placed four lanes into approach flares without demonstrating how they developed or avoided the trench. Its footpaths and crossing rectangles did not establish a connected route around the portals. R1 draws upstream pocket feeds, explicit receiving connections and a connected walking network on the deck side of the interpreted portal edges. Every proposed walk-route intersection with the vehicle envelope is marked for signal control.

The old future deck was drawn as if its width and alignment had been reserved structurally. R1 shows only an airspace investigation corridor with no approved supports, ramps, soffit, foundations or load path. The old projected-coordinate GeoJSON used a legacy CRS member. R1 exports standard longitude/latitude GeoJSON and a separate metre-based DXF, both from the same feature register.

5. Candidate A lane allocation and receiving-lane rule

Candidate A retains two through lanes in each surface direction as a scenario for testing. This is a continuity choice, not a conclusion that two lanes are the correct final allocation. Each through lane has a persistent identity: T1 is the kerbside lane and T2 the median-side lane. T1 connects to T1 and T2 to T2 at the far end. No third through lane is added at a signal and then dropped at its exit.

The four origin approaches each receive a separate left-turn path and right-turn path. Left-turn channels separate upstream of the main through/right throat. Their pedestrian interfaces are signal-controlled; none is a free left turn. Right-turn pockets feed movement-specific connectors, including the north-origin right turn at the southern terminal and the south-origin right turn at the northern terminal. These route choices must be signed before the pocket feed, not left for drivers to discover on the bridge deck.

Northbound and southbound traffic therefore preselects its eventual turn before entering the compound junction. A driver bound for the downstream right turn does not change lanes between the two surface crossings. The right-turn feeder remains a separate route across that short connecting section. The main through lanes remain available for straight movement.

Turning routes terminate in identified receiving lanes: WN and EN use NB-T1; ES and WS use SB-T1; NE uses EB-T1; SW uses WB-T1; NW uses WB-T2; SE uses EB-T2. Different movements can share the same receiving lane only under separate stages, after clearance. A dedicated connector does not mean an extra permanent exit lane for every origin. The geometry does not rely on a merge with a simultaneously released through stream.

The northern frontage axis moves 3.5 m outward and the southern axis 8.0 m outward from the two mapped junction averages. Their proposed local reference separation is 48.044 m. These offsets were chosen to keep the drawn envelopes clear of the interpreted open trenches. They are proposed design changes, not corrected measurements of the existing layout, and need a full alignment and footprint study.

6. Dimensions, pockets and operating environment

The drawing uses a 3.25 m lane-envelope target, a 2.00 m median allocation, a 0.50 m turn-pocket separator allowance, a 2.50 m unobstructed walking target and 4.00 m crossing markings. The central allocation diagram shows six lanes, two separator allowances and a median, totalling 22.50 m before footpaths, parapets, edge clearance and utility zones. These are explicit design assumptions; this package does not claim they satisfy every applicable Indian road or bridge requirement.

The lane curves are editable centreline hypotheses surrounded by width envelopes. They are not tangent/spiral/circular-arc horizontal alignment designs. Offset curves, corner transitions and tie-ins need engineering regularisation. The available width normal to the path, kerb shy distance, vehicle overhang and curve widening must be checked along the complete alignment, particularly in tapers and at skewed crossings.

The proposed surface environment is 30 km/h, with low-speed turning to be tested at approximately 10-15 km/h. These are study objectives, not measured speeds, a speed-limit order or proof that the geometry achieves those speeds. Approach operating speeds, visibility and stopping requirements may alter the transition lengths and island noses.

Pocket feeds are drawn upstream. The north/south right-pocket storage illustrations are about 30 m and the frontage right pockets about 65-70 m; left pockets are shorter and follow curved entries. The movement register records geometric full-width lengths and the limits used to calculate them. These lengths are not demand-based recommendations. Required storage, taper length and deceleration allowance remain unset. Use measured 95th-percentile queues, arrival variation, heavy-vehicle composition, pocket blockage and the adopted design method to size them. Test overflow beyond the pocket in microsimulation.

The road envelope extending beyond the image-traced pavement is shown as a footprint hold. It is neither a land-acquisition quantity nor proof that the area is available. The union of vehicle, footpath, furniture and drainage requirements must fit confirmed ownership and structural constraints. Preserve pedestrian width when

comparing alternatives. If the two-through-lane scenario cannot pass that test, redraw and test a constrained lane scenario or a different movement strategy.

7. Islands and lane-change control

Raised-island candidates are taken from the interpreted surface area after excluding the lane screening envelopes. This removes the earlier practice of drawing kerbs through turning movements. Non-mountable treatment is intended where the final design demonstrates adequate vehicle and pedestrian clearances. Kerb profile, island nose geometry, impact exposure, reflectors, drainage openings and structural anchorage remain to be designed.

Physical separation can lock lane choice on approaches and on portions of the connecting section that are outside turning sweeps. It cannot continue uninterrupted across every crossing trajectory at a signalised junction. Within those conflict areas, persistent lane guidance, exclusive movement stages, signs and enforcement must carry the lane assignment. The concept therefore implements a no-lane-change operating rule; it does not falsely promise a continuous physical barrier around every moving vehicle.

Do not treat every small leftover polygon as a usable refuge or a buildable concrete island. Slivers are screened out, and pedestrian routes cut through selected island areas. A refuge needs at least the adopted 2.0 m clear depth target plus adequate capacity, accessible geometry and clearance from vehicle overhang. If those conditions cannot be provided, combine the pedestrian exposure into one crossing or revise the approach allocation.

8. Signal operation and queue containment

The proposed operating concept treats the two terminals as one compound junction with coordinated detection and controller logic. There is no planned queue storage between them. Entry is permitted only when the complete route, receiving lane and relevant exit can accept the released traffic. Secondary signal heads, stop-line locations and detector positions require a controller and human-factors design so drivers receive an unambiguous continuous route through the compound area.

The illustrated stage sequence is A, B, C, D, E, P. A serves both crossroad through directions and their near-side left turns. B serves the north-origin right to west. C serves the south-origin right to east. D serves west-origin through and both turns. E serves east-origin through and both turns. P holds all surface vehicle movements for pedestrian crossings. The underpass traffic remains outside these surface signal stages.

B and C are separate because the right turns at opposite terminals cross the other direction's connecting path. Similarly, the two frontage right turns are not assumed compatible just because they originate on different sides. The machine-readable conflict matrix screens intersections of 2.8 m moving-object envelopes inside the entry gates. A clear cell in that matrix is not a certified controller conflict definition or a swept-path result.

The six stages are a conservative starting logic, not an optimised timing plan. They may impose unacceptable delay or long pedestrian waits. Counts, queue data and pedestrian demand may support different compatible overlaps or require a different arrangement. Every stage change needs measured-speed-based yellow and all-red clearance. The last admitted vehicle must clear the full conflict route before the next conflicting release. Pedestrian clearance must be protected from every turning stream.

No cycle length, green split, intergreen, offset, queue threshold or assumed capacity is supplied. A finite clearance distance and travel time through the compound junction must be measured for each design vehicle. Provide downstream occupancy detection, entry inhibition, queue-tail alarms, controller fault behaviour and an emergency operating plan. Bus priority and emergency pre-emption must retain clearance and pedestrian protection, including safe recovery to normal coordination.

The candidate contains no U-turn connectors. This is not a claim that existing U-turns are prohibited or unnecessary. Survey legal movements, actual use and bus routes. If a required U-turn cannot be accommodated elsewhere with acceptable access, revise this movement plan before adoption.

9. Walking, accessibility and transit

The walking concept connects the northern and southern crossroad footpaths, all four surface frontage approaches and the two deck-side paths. The four outer corner paths follow the outside of the left-turn channels. Direct connections to the deck-side paths cross those channels under the P stage. This avoids routing a pedestrian along a slip-lane carriageway or onto the open trench.

Each crossing is 4.0 m wide in the drawing. Vehicle-envelope intersections are split only where a nominal refuge gap is at least 2.0 m; shorter gaps are included in one crossing. The resulting exposure lengths, roughly 3-11 m in this version, are in crossing_schedule.json. These are generated geometric lengths, not a pedestrian timing design. Refuge capacity, visibility and effective clear depth remain unverified, so P clearance must cover the full route unless safe staged operation is demonstrated.

Provide accessible signals, tactile information, flush transitions, drainage and lighting. Maintain the 2.5 m walking zone independently of poles, trees, sign foundations, utilities, queues and vendors. Confirm applicable accessibility requirements, crossfalls and ramp details in the Indian guidance and with an accessibility audit. Do not impose an overbridge detour as a substitute for completing these surface connections.

Four labelled bus-stop investigation sites sit beyond the immediate compound area on the departure corridors. They are not verified existing stops, approved relocations or designed bus bays. Confirm routes, boarding and alighting counts, dwell distributions, stop spacing, bus bunching and passenger transfer desire lines with BMTC. Compare in-lane stopping, a stopping lane or a bay based on delay and safety. A curbside stop that blocks T1 temporarily changes the usable receiving-lane supply; model that explicitly and locate detectors accordingly.

All sixteen lane connectors are intended to be assessed for the selected BMTC and emergency design vehicles. The study has not demonstrated their swept paths. Obtain vehicle length, width including mirrors, wheelbase, front and rear overhang, steering limits and turning behaviour. Test the relevant rigid buses, any articulated bus intended for the corridor, fire appliances, ambulances and the appropriate permitted freight combination. Include two-way passing, rear-wheel offtracking, body swing, island clearance, stopped vehicles and emergency access. Required freight movements must inform the design vehicle, not an assumed passenger-car path.

10. Existing and future structures

The interpreted open trenches and portal edges remain exclusion features. Automated drawing checks find no proposed lane or walking envelope overlapping those interpreted open trench polygons. That statement is restricted to the tracing and does not certify a structural separation distance. Keep the underpass and its portals in operation conceptually; do not infer an extension of the existing deck from the white road polygons.

Obtain as-built deck limits, retaining-wall geometry, foundations, bearing and expansion-joint positions, reinforcement, waterproofing, load ratings, drainage and utilities. Check every raised island, parapet alteration, signal foundation and footpath load. Avoid drilling, anchors, planting pits or heavy fill on the slab until structural approval. Check flood routing, inlet blockage, monsoon discharge and underpass emergency access.

Future CityWays provision is a 12 m overall airspace study corridor following the crossroad. It is a planning placeholder, not a chosen deck cross-section. No surface land is exclusively allocated to its supports. The two bus/HOV running lanes, barriers, evacuation provision, maintenance access, loading, grades, headroom and ramps require a separate feasibility study. A planning support-exclusion polygon combines the interpreted trenches and deck with a 5 m screening buffer; the 5 m is an assumption, not a foundation setback standard.

Investigate whether a long-span arrangement can place foundations outside the existing structure's influence and preserve all required surface lanes, refuges and footpaths. Locate any ramps beyond the junction and queue influence areas only after a corridor study. If that cannot be achieved, change or abandon the future alignment. Do not narrow the surface concept now on the assumption that an elevated structure will solve its capacity problem later.

11. Surveys and counts required before engineering design

- Topographic control and levels: survey kerbs, carriageway edges, markings, islands, portals, parapets, steps, ramps, drains, utilities, trees, access points and sight obstructions. Extend beyond all candidate tie-ins and maximum observed queue tails. Tie the work to the authority's accepted horizontal and vertical datum.

- Property and utilities: confirm public road reserve, parcels, easements and access rights; obtain utility records and verify critical crossings and foundation locations with appropriate detection and investigation.

- Structural investigation: obtain and reconcile as-builts, inspect deck and walls, map joints and waterproofing, determine permissible loads and foundation influence zones, and verify underpass clearance and drainage.

- Classified turning counts: collect simultaneous counts at both terminals and all slip entries/exits, including U-turns, motorcycles, autorickshaws, cars, buses, goods vehicles, cycles and pedestrians. Retain short-interval data, such as 15-minute bins, and vehicle trajectories where permitted. Start with representative weekdays and a weekend day covering AM, midday and PM conditions; extend the sample where day-to-day or seasonal variability requires it. This is a sampling proposal, not a statutory duration.

- Queues and travel times: record lane-by-lane queue length by cycle, blocking duration, discharge headways, arrivals, residual queues, maximum and percentile queues, corridor travel times, diversion and downstream receiving occupancy. Include wet-weather and atypical high-demand checks.

- Signals: log existing controller plans, actual phase sequence, detector operation, intergreens, cycle variation, failures, coordination, signal visibility and red-light behaviour. Any archived timing sheet must be verified against current operation.

- Walking and cycling: count desire lines, waiting times, crossing times, mobility-aid users, school and workplace peaks, cycle movements, refuge crowding and informal crossings. Observe pedestrian access to every bus stop and adjoining frontage.

- BMTC and emergency services: obtain route and stop records, bus dimensions, dwell/bunching data and transfer patterns; confirm emergency vehicle dimensions, response routes and pre-emption requirements. Establish the governing freight vehicles and time restrictions with the responsible authority.

- Safety and environment: review available geocoded crash history and video conflicts, speeds and sight distance; audit lighting, personal security, stormwater, heat/shade and maintenance. Use qualified road-safety and accessibility reviewers independent of the design team.

12. Simulation and design decision process

Build and validate an existing-condition network first. Use observed lane use and motorcycle behaviour rather than a default homogeneous-lane model. Calibrate saturation flow, desired speeds, turning speeds, gap acceptance, lane-changing behaviour, bus dwell and compliance. Validate on a different observed period or day using queues and journey times as well as counts. State the adopted fit criteria before tuning and retain the input data and random seeds.

Compare the existing operation, a minimal channelisation treatment, Candidate A, and at least one constrained-footprint lane/movement alternative. Consider a single-node or other junction arrangement only after showing how it fits pedestrian and structural requirements. Run multiple seeds and peak, off-peak, queue spillback, bus bunching, blocked-lane, incident and emergency scenarios. Demand-growth scenarios must be labelled assumptions and traced to a source or policy case.

Report person-delay, bus reliability and passenger delay, pedestrian wait and exposure, discharge throughput, stops, cycle failures, 95th-percentile queues, maximum spillback and approach-to-exit lane utilisation. Separately audit safety; simulated travel-time improvement is not proof of safety. Include sensitivity to imperfect lane discipline, long vehicles, uneven arrivals and unavailable downstream space.

Approve a final arrangement only after the footprint, accessibility, swept paths, structural capacity and operating performance all pass review. Then develop surveyed alignments, kerbs, drainage, levels, pavement, structures, lighting, signs, markings, signals, utilities, staging and temporary traffic management. The six-stage sketch is not to be loaded into a controller as supplied.

13. Validation performed and remaining holds

The generator checks geometry validity, interpreted trench intrusion, raised-island/lane overlap, unmarked walking-centreline exposure and same-stage 2.8 m envelope intersections inside the entry gates. The DXF is reopened and audited. GeoJSON is checked for longitude/latitude bounds and feature identifiers. Both CAD and GIS originate from one feature register. The sheets and report are visually inspected after rendering.

These checks establish file integrity and internal drawing consistency only. They do not validate vehicle swept paths, signal conflict-monitor programming, lane capacity, structural clearance, gradients, drainage, road reserve, accessibility details or safety performance. Residual fit uncertainty and candidate widening areas are shown, not removed from the data. The final validation report distinguishes machine checks from engineering holds.

14. References and their role

OpenStreetMap copyright and licence. Source for mapping attribution and reuse; road geometry remains a community mapping base rather than a survey. The original and current extract are archived with hashes.

ITDP India, Complete Streets Framework Toolkit. Indian public-realm and transport planning context, including walking, cycling and public transport. Used as general process guidance; the dimensions in this candidate are project assumptions, not claimed quotations from this publication.

ITDP India, Healthy Streets Design Workbook. Identifies the street-design process and public-realm scope to be considered. Final designers should consult the complete workbook and applicable authority specifications.

Government of India, Harmonised Guidelines and Standards for Universal Accessibility in India 2021. Reference for detailed accessibility design and audit. No unverified clause-level compliance or ramp gradient is asserted here.

FHWA, Signalized Intersections: Informational Guide, Chapter 9. Background on pedestrian signal treatments and their delay trade-offs. Its US context and age mean it is supplementary guidance, not an Indian design standard or a source for adopted timings.

FHWA, Signalized Intersections: Informational Guide, Chapter 10 and FHWA, Alternative Intersections/Interchanges, Chapter 9. Retained from the earlier concept for alternative-form discussion. They do not establish that a tight urban diamond or single-point interchange is preferred at this site.

The earlier basis also cited FHWA pedestrian/bicycle interchange guidance and DDA/UTTIPEC guidance. Those legacy URLs are preserved here; their detailed content was not relied upon for this revision. Obtain the current applicable IRC urban-road, intersection, pedestrian, signs, markings and bus-stop guidance and the responsible Bengaluru authority's requirements before engineering design.